Name _____ _____ Baggett/Fall2026/207/Quiz7

first name last name

Score

No Calculators. / 20

Questions 1 and 2. Circle **True** or **False**. Then, in **one sentence**, say why — if the statement is false, the fastest way is usually to give an example that breaks it. **1 point for the circle, 1 for the reason.**

1. (**2** pts) If $f'(x) < 0$ for $1 < x < 6$, then f is decreasing on $(1, 6)$.

True **False**

Why? _____ /2

2. (**2** pts) If f and g are differentiable and $f(x) \geq g(x)$ for $a < x < b$, then $f'(x) \geq g'(x)$ for $a < x < b$.

True **False**

Why? _____ /2

3. (**7** pts) Let $f(x) = x^3 - 3x^2 - 9x + 5$.

(a) Find the critical numbers of f, and classify each one as a local maximum, a local minimum, or neither. **(4 points)**

(b) On what interval(s) is f concave up and concave down? Give the inflection point as an ordered pair. **(3 points)**

/7

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4. (**6** pts) A function f is differentiable everywhere, and

$$f'(x) = (x+3)(x-1) \qquad \text{and} \qquad f''(x) = 2(x+1).$$

You are **not** given a formula for f.

(a) On what intervals is f increasing and decreasing, and what happens at each critical number? **(3 points)**

(b) On what intervals is f concave up and concave down, and where is the inflection point? **(3 points)**

/6

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5. (**3** pts) Suppose $f'(2) = 0$ and $f''(2) = 0$. What does the Second Derivative Test tell you about $x = 2$, and what should you do instead?

/3

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